SEAT NO:- 31470002

DR. HOMI BHABHA STATE UNIVERSITY

INSTITUTE OF SCIENCE

DEPARTMENT OF BUSINESS AI

ACADEMIC YEAR-2026-27

INTRODUCTION TO BUSINESS ANALYTICS – LAB

NAME:- YASH BABAN AGRE

CLASS:- SY.BSC.BUSINESS AI

SEAT NO:- 31470002

SUBJECT:- INTRODUCTION TO BUSINESS ANALYTICS – LAB

SEMESTER:- III

TEACHER SIGN:- _____

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DEPARTMENT OF BUSINESS AI CERTIFICATE

This is to certify that **Mr./Ms.YASH BABAN AGRE** Exam Seat No./Roll No. **No. 31470002** B.Sc. Business AI (Semester III) has completed the practical work in the subject during the academic year 2026-27 under the guidance of **PROF.FARHAT MAM** being the partial requirement for the fulfilment of the curriculum of Degree of Bachelor of Science in Business AI at The Institute of Science, Dr. Homi Bhabha State University.

Professor In-Charge **Coordinator**

Date: - ……. **College seal**

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Practical 1

Aim	To import a raw sales/HR dataset, inspect its structure, identify data quality issues, and produce a clean, analysis-ready dataset.
Procedure	1. Open a new Excel workbook, go to Data tab → Get Data → From File → From Text/CSV, browse to hr_data.csv and click Import. 2. In the preview window, check File Origin is UTF-8, Delimiter is Comma and each column header is in its own column; click Load and rename the worksheet tab "Raw_Data". 3. Press Ctrl + Shift + End to check the dataset shape, check the Number Format of each column on the Home tab, freeze the top row and confirm the row count with =COUNTA(A2:A501). 4. Select MonthlyIncome and Name → Conditional Formatting → Highlight Cells Rules → More Rules → Format only cells that contain → Blanks → yellow fill, to find missing values. 5. Select the full data range → Conditional Formatting → Highlight Duplicate Values to preview duplicate rows. 6. Calculate Q1 and Q3 with =QUARTILE.INC(MonthlyIncome_range,1) and (…,3), then IQR, Lower Bound = Q1 - 1.5*IQR and Upper Bound = Q3 + 1.5*IQR; apply =OR(B2<Lower_Bound,B2>Upper_Bound) in red. 7. Remove duplicates (Data → Remove Duplicates, all columns); fill missing income with the department median using =MEDIAN(IF(Department_range="Sales",MonthlyIncome_range)) (Ctrl+Shift+Enter); flag the missing Name without deleting the row. 8. Standardize text with =PROPER(Department_cell) and =UPPER(Gender_cell), then Paste Special → Values; fix JoiningDate using Data → Text to Columns → Delimited → Date: YMD. 9. Save the cleaned file as hr_data_cleaned.xlsx. 10. Create a new worksheet tab named "Cleaning Log" and record each action in a table of Step, Action Taken and Reason.
Output	**Question 1 — Import hr_data.csv into Excel and inspect how the data loads.** **Input Data — hr_data.csv (sample of 12 rows shown):**

EmployeeID	Name	Department	Age	Gender	MonthlyIncome	JoiningDate	AttritionFlag
E001	Aarav Shah	sales	29	M	42000	12-03-2021	No
E002	Priya Nair	Sales	34	F		05-07-2019	No
E003	Rohit Verma	HR	41	M	55000	22-11-2018	Yes
E004	Rohit Verma	HR	41	M	55000	22-11-2018	Yes
E005	Sneha Iyer	finance	26	F	38000	15-01-2022	No
E006	Kunal Mehta	Finance	38	M	950000	30-06-2017	No
E007		Marketing	31	F	47000	09-09-2020	No
E008	Divya Rao	sales	45	F	61000	14-02-2016	Yes
E009	Arjun Singh	IT	27	M	52000	2021/08/19	No
E010	Meena Kumar	Marketing	33	F	44000	03-12-2019	No
E011	Aditya Joshi	IT	29	M		25-04-2021	No
E012	Kavya Reddy	finance	36	F	49000	11-10-2018	No

Import Result:

Item	Value
File Origin	UTF-8
Delimiter	Comma
Worksheet tab name	Raw_Data
Worksheet size	500 rows (plus header) × 8 columns

Question 2 — Inspect the dataset's shape, column data types, and a sample of records.

Dataset Shape:

Check	Result
Used range (Ctrl + Shift + End)	A1:H501
Rows (excluding header)	500
Columns	8
=COUNTA(A2:A501)	500

Column Data Types (Number Format on Home tab):

Column	Number Format Shown	Status
EmployeeID	General / Text	OK
Name	General / Text	OK
Department	General / Text	OK
Age	General / Number	OK
Gender	General / Text	OK
MonthlyIncome	General / Number	OK
JoiningDate	Date (E009 value 2021/08/19 shows General / Text)	Data type problem
AttritionFlag	General / Text	OK

Question 3 — Identify missing values, duplicate rows, and potential outliers in MonthlyIncome.

Missing Values (highlighted yellow):

Column	EmployeeID	Cell
MonthlyIncome	E002	Blank
MonthlyIncome	E011	Blank
Name	E007	Blank

Duplicate Rows (highlighted):

EmployeeID	Name	Department	Age	Gender	MonthlyIncome	JoiningDate	AttritionFlag
E003	Rohit Verma	HR	41	M	55000	22-11-2018	Yes
E004	Rohit Verma	HR	41	M	55000	22-11-2018	Yes

Outlier Calculation (10 recorded MonthlyIncome values in the sample):

Cell	Formula	Value
Q1	=QUARTILE.INC(MonthlyIncome_range,1)	₹44,750
Q3	=QUARTILE.INC(MonthlyIncome_range,3)	₹55,000
IQR	=Q3_cell - Q1_cell	₹10,250
Lower Bound	=Q1_cell - 1.5*IQR_cell	₹29,375

Upper Bound	=Q3_cell + 1.5*IQR_cell	₹70,375

Outliers flagged in red:

EmployeeID	MonthlyIncome	Condition
E006	₹950,000	> Upper Bound (₹70,375)

Question 4 — Clean and standardize the dataset.

Cleaned Dataset (sample) — hr_data_cleaned.xlsx:

EmployeeID	Name	Department	Age	Gender	MonthlyIncome	JoiningDate	AttritionFlag
E001	Aarav Shah	Sales	29	M	42000	12-03-2021	No
E002	Priya Nair	Sales	34	F	46500	05-07-2019	No
E003	Rohit Verma	HR	41	M	55000	22-11-2018	Yes
E005	Sneha Iyer	Finance	26	F	38000	15-01-2022	No
E006	Kunal Mehta	Finance	38	M	950000	30-06-2017	No
E007	(missing - flagged)	Marketing	31	F	47000	09-09-2020	No
E008	Divya Rao	Sales	45	F	61000	14-02-2016	Yes
E009	Arjun Singh	IT	27	M	52000	19-08-2021	No
E010	Meena Kumar	Marketing	33	F	44000	03-12-2019	No
E011	Aditya Joshi	IT	29	M	48000	25-04-2021	No
E012	Kavya Reddy	Finance	36	F	49000	11-10-2018	No

Question 5 — Document each cleaning step taken, in a log a manager could review.

Data Cleaning Log (worksheet "Cleaning Log"):

Step	Action Taken	Reason
1	Removed 1 duplicate row (E004, duplicate of E003)	Same EmployeeID/record appeared twice due to export error
2	Filled missing income for E002, E011 with department median	Avoids distortion from extreme values vs. mean
3	Flagged E007 for missing Name; retained row	No reliable way to infer name, but other fields are valid
4	Standardized Department/Gender text casing	Prevents "sales" and "Sales" being treated as different categories
5	Converted JoiningDate to a single consistent date format	Enables tenure calculations later
6	Flagged E006 income (₹950,000) for HR verification	Possible data-entry error; not deleted pending confirmation

Practical 2

Aim	To compute and interpret central tendency, dispersion, and distribution-shape statistics on business data.
Procedure	1. Enter the MonthlyIncome values from hr_data_cleaned.xlsx in one column (MonthlyIncome_range). 2. Label a cell "Mean" and enter =AVERAGE(MonthlyIncome_range); label the next cell "Median" and enter =MEDIAN(MonthlyIncome_range); label the next cell "Mode" and enter =MODE.SNGL(MonthlyIncome_range). 3. Format all result cells as currency (₹) using Home → Number Format → Currency. 4. Label a cell "Variance" and enter =VAR.S(MonthlyIncome_range); label a cell "Std Dev" and enter =STDEV.S(MonthlyIncome_range). 5. Label a cell "Range" and enter =MAX(MonthlyIncome_range)-MIN(MonthlyIncome_range). 6. Label cells "Q1", "Q2 (Median)" and "Q3" and enter =QUARTILE.INC(MonthlyIncome_range,1), (…,2) and (…,3) respectively. 7. Label a cell "IQR" and enter =Q3_cell - Q1_cell. 8. Label cells "Skewness" and "Kurtosis" and enter =SKEW(MonthlyIncome_range) and =KURT(MonthlyIncome_range). 9. Compile all results into one summary table on a new worksheet tab named "Descriptive Summary".
Output	**Input Data — MonthlyIncome column (12 values):**

EmployeeID	MonthlyIncome
E001	42,000
E002	46,500
E003	55,000
E005	38,000
E006	95,000
E008	61,000
E009	52,000
E010	44,000
E011	48,000
E012	49,000
E013	53,000
E014	41,000

Question 1 — Compute mean, median, and mode of MonthlyIncome.

Statistic	Formula	Value
Mean	=AVERAGE(MonthlyIncome_range)	₹52,042
Median	=MEDIAN(MonthlyIncome_range)	₹48,500
Mode	=MODE.SNGL(MonthlyIncome_range)	N/A (no exact repeat in sample)

Question 2 — Compute variance, standard deviation, and range of MonthlyIncome.

Statistic	Formula	Value
Variance	=VAR.S(MonthlyIncome_range)	225,111,742
Std Dev	=STDEV.S(MonthlyIncome_range)	₹15,004
Range	=MAX(MonthlyIncome_range)-MIN(MonthlyIncome_range)	₹95,000 − ₹38,000 = ₹57,000

Question 3 — Compute quartiles and the Interquartile Range (IQR).

Statistic	Formula	Value
Q1	=QUARTILE.INC(MonthlyIncome_range,1)	₹43,500
Q2 (Median)	=QUARTILE.INC(MonthlyIncome_range,2)	₹48,500
Q3	=QUARTILE.INC(MonthlyIncome_range,3)	₹53,500
IQR	=Q3_cell - Q1_cell	₹10,000

Question 4 — Compute skewness and kurtosis of MonthlyIncome.

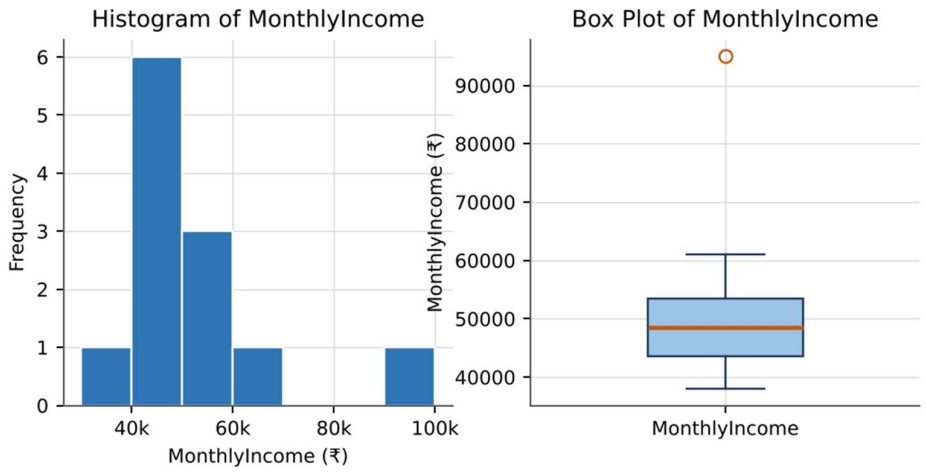
Statistic	Formula	Value
Skewness	=SKEW(MonthlyIncome_range)	+2.39
Kurtosis	=KURT(MonthlyIncome_range)	+6.79

Question 5 — Interpret all statistics together in a business summary.

Statistical Summary Report (worksheet "Descriptive Summary"):

Statistic	Value	Business Meaning
Mean	₹52,042	Average pay across sampled employees
Median	₹48,500	Pay of the "typical" employee
Std Dev	₹15,004	Degree of pay variation
IQR	₹10,000	Spread of the middle-earning 50%
Skewness	+2.39	A few high earners pulling distribution right

Chart:



Histogram data (bin width ₹10,000):

Bin (₹)	Frequency
30,000–40,000	1
40,000–50,000	6
50,000–60,000	3

60,000–70,000	1
70,000–80,000	0
80,000–90,000	0
90,000–100,000	1

Chart Description: The histogram of MonthlyIncome (bins of ₹10,000 from ₹30,000 to ₹100,000) is right-skewed: 1 value in ₹30,000–40,000, 6 in ₹40,000–50,000, 3 in ₹50,000–60,000, 1 in ₹60,000–70,000 and 1 in ₹90,000–100,000 (none in ₹70,000–90,000), with a long tail to ₹95,000. The box plot shows the median at ₹48,500, the box from Q1 ₹43,500 to Q3 ₹53,500, whiskers from ₹38,000 to ₹61,000 (fences ₹28,500 and ₹68,500) and a single outlier at ₹95,000.

Practical 3

Aim	To summarize sales data by region and product-year combinations using pivot tables, and extract business insights.
Procedure	1. Click anywhere inside the data range, go to Insert tab → PivotTable → New Worksheet → OK. 2. In the Field List, drag Region into the Rows box and SalesAmount into the Values box (confirm "Sum of SalesAmount"); rename the pivot sheet tab "Sales by Region". 3. Insert a second PivotTable from the same source data range; drag Product to Rows, Year to Columns and SalesAmount to Values (Sum); rename the pivot sheet tab "Sales by Product-Year". 4. Click inside the "Sales by Region" pivot table → PivotTable Analyze tab → Insert Slicer, tick Product and click OK. 5. Drag the slicer (buttons "Electronics" and "Apparel") to a clear area next to the pivot table and click "Electronics" to confirm the pivot table updates. 6. Click inside the "Sales by Region" pivot table → PivotTable Analyze tab → PivotChart → Clustered Column Chart → OK. 7. Click the chart title placeholder and type "Regional Sales Performance". 8. Right-click the Product slicer → Report Connections → tick the new PivotChart's underlying pivot table so the slicer also filters the chart. 9. Derive 3 business insights from the pivot tables created above.
Output	**Input Data — sales_data.csv (sample of 14 rows shown):**

OrderID	Region	Product	Year	Month	SalesAmount	AdSpend
S101	North	Electronics	2023	Jan	125,000	18,000
S102	North	Apparel	2023	Jan	64,000	9,000
S103	South	Electronics	2023	Jan	58,000	7,000
S104	East	Apparel	2023	Feb	71,000	9,500
S105	West	Electronics	2023	Feb	98,000	14,000
S106	North	Electronics	2024	Jan	142,000	21,000
S107	South	Apparel	2024	Jan	49,000	6,000
S108	South	Electronics	2024	Feb	51,000	6,500
S109	East	Electronics	2024	Feb	87,000	12,000
S110	North	Apparel	2024	Mar	69,000	9,800
S111	West	Apparel	2023	Mar	56,000	7,200
S112	West	Electronics	2024	Mar	102,000	15,500
S113	East	Apparel	2024	Apr	73,000	9,700
S114	South	Electronics	2023	Apr	53,000	6,800

Question 1 — Build a pivot table showing total Sales by Region.

Pivot Table (worksheet "Sales by Region"):

Region	Sum of SalesAmount
East	231,000
North	400,000
South	211,000

West	256,000
Grand Total	**1,098,000**

Question 2 — Build a pivot table showing Sales by Product × Year.

Pivot Table (worksheet "Sales by Product-Year"): Sum of SalesAmount

Product	2023	2024	Grand Total
Apparel	191,000	191,000	382,000
Electronics	334,000	382,000	716,000
Grand Total	**525,000**	**573,000**	**1,098,000**

Question 3 — Add slicers and filters to the Region pivot table.

Slicer "Product" — buttons: Electronics | Apparel. Pivot table after clicking "Electronics":

Region	Sum of SalesAmount
East	87,000
North	267,000
South	162,000
West	200,000
Grand Total	**716,000**

Question 4 — Create a pivot chart from the Region pivot table.

Chart data (Clustered Column Chart):

Region	Sum of SalesAmount
East	231,000
North	400,000
South	211,000
West	256,000

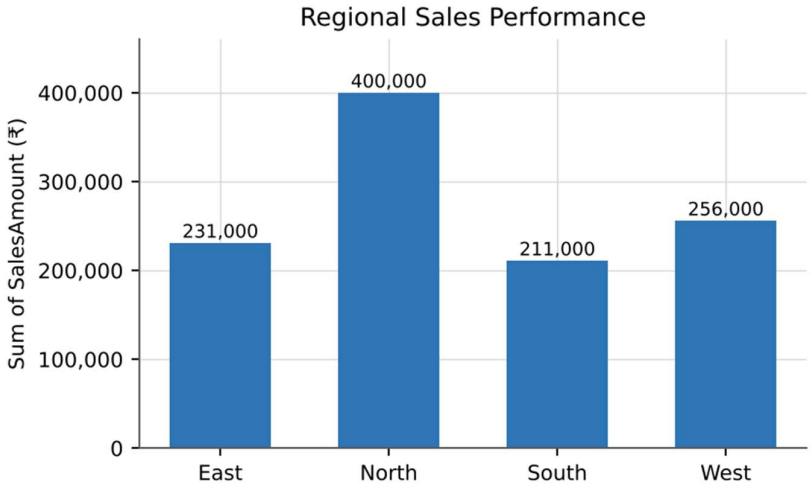


Chart Description: Clustered column chart titled "Regional Sales Performance". The X-axis shows the regions East, North, South and West; the Y-axis shows Sum of SalesAmount from 0 to 450,000. A single series has four columns: East 231,000, North 400,000 (tallest), South 211,000 (shortest) and West 256,000. With the Product slicer connected, selecting "Electronics" changes the columns to East 87,000, North 267,000, South 162,000 and West

	200,000. **Question 5 — Derive 3 business insights from the pivot tables created above.** **Business Insights:** 1. **Region concentration:** North contributes the highest total sales (₹400,000 in the sample) — priority market for inventory allocation. 2. **Product-Year trend:** Electronics grew from ₹334,000 (2023) to ₹382,000 (2024), while Apparel stayed flat at ₹191,000 both years — apparel line may need refreshing. 3. **Underperforming combination:** South's electronics sales (₹58,000 + ₹51,000 + ₹53,000 = ₹162,000 across 3 orders) have the lowest average order value among regions for that category (₹54,000, against ₹133,500 in North, ₹100,000 in West and ₹87,000 in East) — signals a distribution or marketing gap in South.

Practical 4

Aim	To create and interpret standard business charts for communicating data patterns.
Procedure	1. Use the Region summary table from Practical 3, Question 1 (or build one with =SUMIF(Region_range,"North",SalesAmount_range) for each region); select the two-column table → Insert tab → Charts group → Clustered Column Chart. 2. Click Chart Title and type "Total Sales by Region"; click + (Chart Elements) → Axis Titles and label the Y-axis "Sales Amount (₹)"; use Chart Design → Change Colors to apply a consistent blue theme. 3. Build a Month vs. Total Sales table using a Pivot Table (Month in Rows, Sum of SalesAmount in Values), sorted Jan, Feb, Mar, Apr; select it → Insert → Line Chart with Markers; title it "Monthly Sales Trend" and label the X-axis "Month" and Y-axis "Sales Amount (₹)". 4. Build a Product vs. Total Sales table (Electronics, Apparel); select it → Insert → Pie Chart (or Doughnut Chart); right-click the chart → Add Data Labels → More Options → tick Percentage, untick Value; title it "Product Share of Total Sales". 5. Select the AdSpend and SalesAmount columns together (Ctrl for non-adjacent columns) → Insert → Scatter (X,Y) Chart. 6. Click + → Axis Titles and label the X-axis "Ad Spend (₹)" and Y-axis "Sales Amount (₹)"; right-click any data point → Add Trendline → Linear → tick Display R-squared value on chart. 7. Select the Region and SalesAmount columns → Insert → Insert Statistic Chart → Box and Whisker; Excel groups the data by Region and plots quartiles, median line and outlier points; title it "Sales Distribution by Region". 8. Note the best use of each chart type in a business context (Bar/Column, Line, Pie/Donut, Scatter, Boxplot).
Output	**Input Data — sales_data.csv (same sample dataset as Practical 3):**

OrderID	Region	Product	Year	Month	SalesAmount	AdSpend
S101	North	Electronics	2023	Jan	125,000	18,000
S102	North	Apparel	2023	Jan	64,000	9,000
S103	South	Electronics	2023	Jan	58,000	7,000
S104	East	Apparel	2023	Feb	71,000	9,500
S105	West	Electronics	2023	Feb	98,000	14,000
S106	North	Electronics	2024	Jan	142,000	21,000
S107	South	Apparel	2024	Jan	49,000	6,000
S108	South	Electronics	2024	Feb	51,000	6,500
S109	East	Electronics	2024	Feb	87,000	12,000
S110	North	Apparel	2024	Mar	69,000	9,800
S111	West	Apparel	2023	Mar	56,000	7,200
S112	West	Electronics	2024	Mar	102,000	15,500
S113	East	Apparel	2024	Apr	73,000	9,700
S114	South	Electronics	2023	Apr	53,000	6,800

Question 1 — Create a bar/column chart comparing total sales by Region.

Chart data (Region summary table):

Region	Sum of SalesAmount
East	231,000
North	400,000
South	211,000
West	256,000

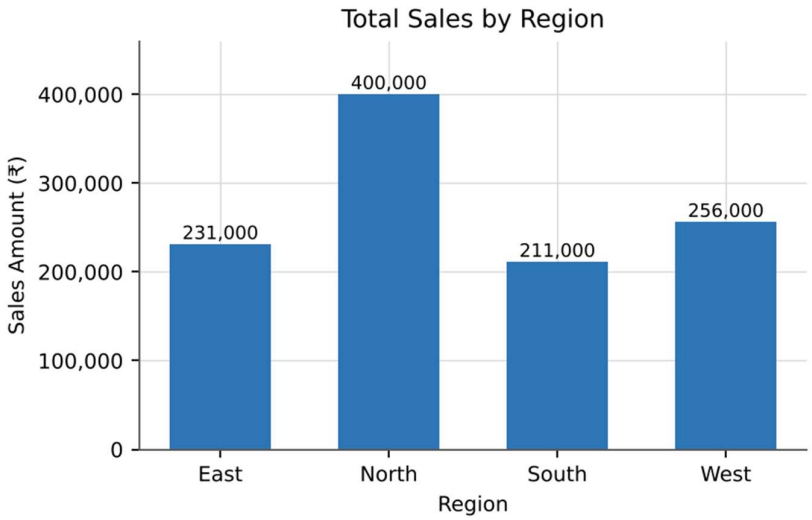


Chart Description: Clustered column chart titled “Total Sales by Region” in a blue theme. X-axis: Region (East, North, South, West). Y-axis titled “Sales Amount (₹)”, scale 0 to 450,000. One series with column heights East 231,000, North 400,000, South 211,000 and West 256,000. North is the tallest column and South the shortest.

Question 2 — Create a line chart showing monthly sales trend.

Chart data (trend table):

Month	Sum of SalesAmount
Jan	438,000
Feb	307,000
Mar	227,000
Apr	126,000

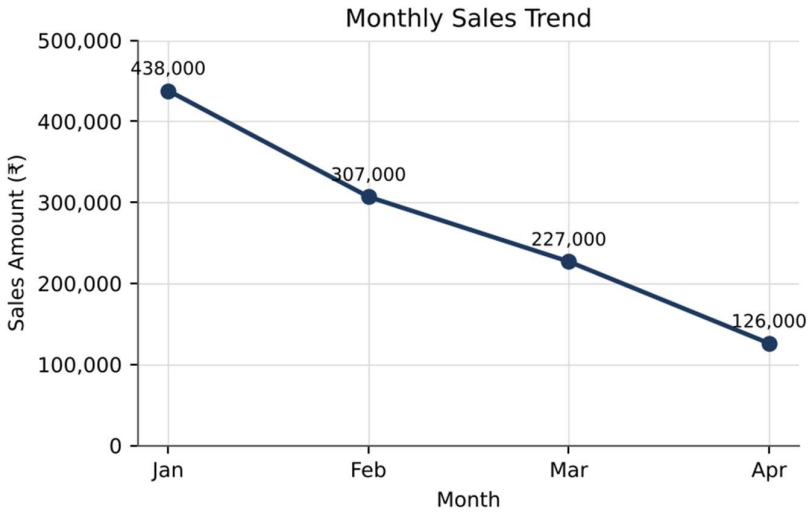


Chart Description: Line chart with markers titled “Monthly Sales Trend”. X-axis titled “Month” (Jan, Feb, Mar, Apr in chronological order); Y-axis titled “Sales Amount (₹)”. The line falls at every step: Jan 438,000 → Feb 307,000 (−131,000) → Mar 227,000 (−80,000) → Apr 126,000 (−101,000).

Question 3 — Create a pie/donut chart showing each product’s share of total sales.

Chart data (Product vs. Total Sales):

Product	Total Sales	Share
Electronics	716,000	65.2%
Apparel	382,000	34.8%
Total	**1,098,000**	**100.0%**

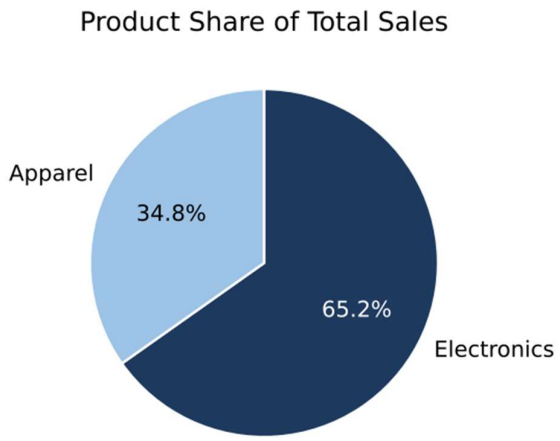


Chart Description: Pie chart titled “Product Share of Total Sales” with percentage data labels (values unticked). Two slices: Electronics 65.2% (₹716,000) and Apparel 34.8% (₹382,000).

Question 4 — Create a scatter plot for the relationship between Advertising Spend and Sales.

Chart data (AdSpend and SalesAmount columns):

OrderID	AdSpend (₹)	SalesAmount (₹)
S101	18,000	125,000
S102	9,000	64,000
S103	7,000	58,000
S104	9,500	71,000
S105	14,000	98,000
S106	21,000	142,000
S107	6,000	49,000
S108	6,500	51,000
S109	12,000	87,000
S110	9,800	69,000
S111	7,200	56,000
S112	15,500	102,000
S113	9,700	73,000
S114	6,800	53,000

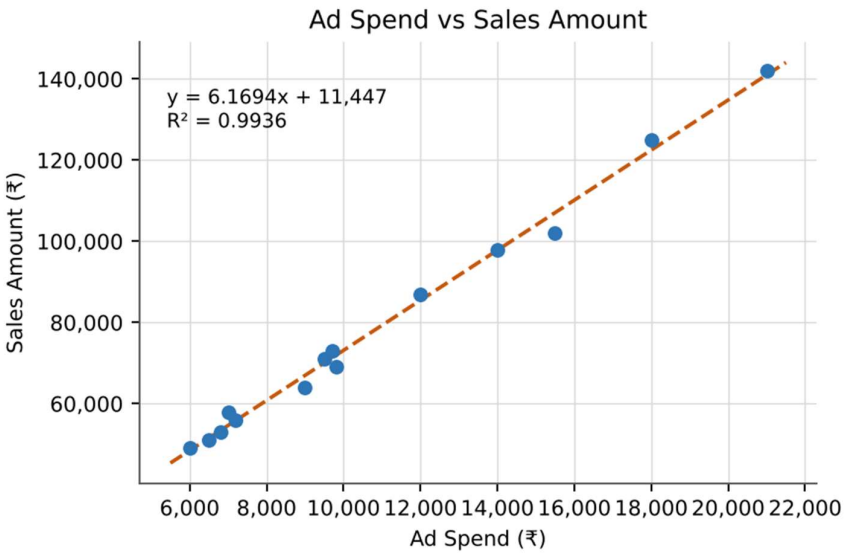


Chart Description: Scatter (X,Y) chart. X-axis titled "Ad Spend (₹)" (6,000 to 21,000); Y-axis titled "Sales Amount (₹)" (49,000 to 142,000). The 14 points rise steadily from the lower left (S107: 6,000 → 49,000) to the upper right (S106: 21,000 → 142,000). The linear trendline displays y = 6.1694x + 11,446.68 and R² = 0.9936.

Question 5 — Create a boxplot for distribution and outliers in Sales by Region.

Chart data (five-number summary by Region; quartiles by Excel's default exclusive-median method):

Region	Minimum	Q1	Median	Q3	Maximum	Mean	Outliers
East	71,000	71,000	73,000	87,000	87,000	77,000	None
North	64,000	66,500	97,000	133,500	142,000	100,000	None
South	49,000	50,000	52,000	55,500	58,000	52,750	None
West	56,000	56,000	98,000	102,000	102,000	85,333	None

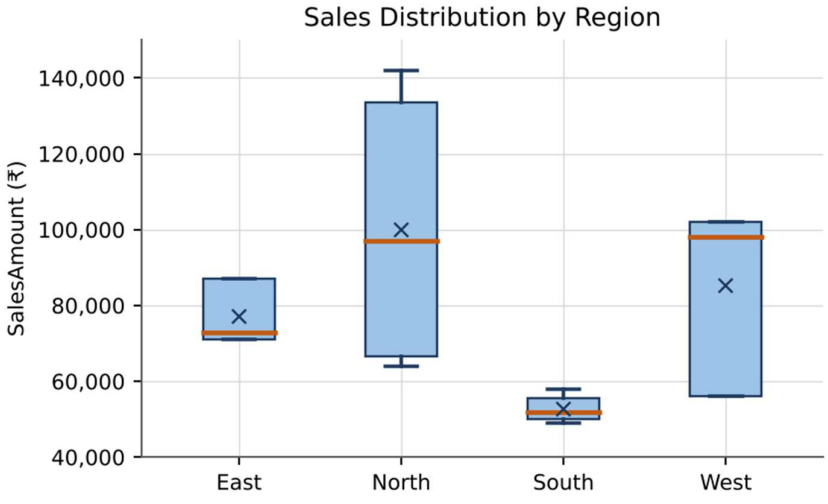


Chart Description: Box and whisker chart titled "Sales Distribution by Region", one box per region (East, North, South, West). North has the widest box (66,500 to 133,500, median 97,000) with whiskers from 64,000 to 142,000. West ranges 56,000 to 102,000 with median 98,000. East is the tightest box (71,000 to 87,000, median 73,000) and South sits lowest (50,000 to 55,500, median 52,000, whiskers 49,000 to 58,000). No outlier points appear in

any region.

Question 6 — Explain when to use each chart type in a business context.

Chart Type	Best Used For
Bar/Column	Comparing totals across discrete categories (e.g., region, product)
Line	Showing trends over a continuous time period (e.g., monthly sales)
Pie/Donut	Showing each category's share of a whole, when categories are few (≤6)
Scatter	Studying relationship/correlation between two numeric variables
Boxplot	Comparing distribution spread and spotting outliers across groups

Practical 5

Aim	To systematically explore a dataset using univariate, bivariate, and grouped analysis to surface actionable insights.
Procedure	1. Enable the Analysis ToolPak: File → Options → Add-ins → Manage: Excel Add-ins → Go → tick Analysis ToolPak → OK. 2. Data tab → Data Analysis → Descriptive Statistics; set Input Range = SalesAmount column, tick Summary Statistics, Output to a new worksheet → OK; repeat for the AdSpend column. 3. For each numeric column, select the column → Insert → Insert Statistic Chart → Histogram. 4. Create the scatter plot from Practical 4, Question 4 and compute =CORREL(AdSpend_range, SalesAmount_range) in a blank cell; repeat for Month-index vs SalesAmount. 5. Data tab → Data Analysis → Correlation; set Input Range to cover SalesAmount and AdSpend, tick Labels in First Row, Output to new location → OK. 6. Select the matrix cells → Home → Conditional Formatting → Color Scales → Red-Yellow-Green scale to create a heatmap. 7. Build a PivotTable with Region and Product both in Rows (Region first, Product nested under it). 8. Drag SalesAmount into Values three times; right-click each → Value Field Settings → set Average, Sum and Count; rename headers "Avg Sales", "Total Sales", "Order Count". 9. Document 5 business insights in an EDA report.
Output	**Input Data — sales_data.csv (continuing with the Practical 3/4 sample dataset):**

OrderID	Region	Product	Year	Month	SalesAmount	AdSpend
S101	North	Electronics	2023	Jan	125,000	18,000
S102	North	Apparel	2023	Jan	64,000	9,000
S103	South	Electronics	2023	Jan	58,000	7,000
S104	East	Apparel	2023	Feb	71,000	9,500
S105	West	Electronics	2023	Feb	98,000	14,000
S106	North	Electronics	2024	Jan	142,000	21,000
S107	South	Apparel	2024	Jan	49,000	6,000
S108	South	Electronics	2024	Feb	51,000	6,500
S109	East	Electronics	2024	Feb	87,000	12,000
S110	North	Apparel	2024	Mar	69,000	9,800
S111	West	Apparel	2023	Mar	56,000	7,200
S112	West	Electronics	2024	Mar	102,000	15,500
S113	East	Apparel	2024	Apr	73,000	9,700
S114	South	Electronics	2023	Apr	53,000	6,800

Question 1 — Perform univariate analysis on each numeric column.

Descriptive Statistics output (Summary Statistics):

Statistic	SalesAmount	AdSpend

Mean	78,428.57	10,857.14
Standard Error	7,699.52	1,244.05
Median	70,000	9,600
Mode	#N/A	#N/A
Standard Deviation	28,808.96	4,654.81
Sample Variance	829,956,043.96	21,667,252.75
Kurtosis	0.3721	0.1623
Skewness	1.0973	1.0308
Range	93,000	15,000
Minimum	49,000	6,000
Maximum	142,000	21,000
Sum	1,098,000	152,000
Count	14	14

Histogram data:

SalesAmount Bin (₹)	Frequency	AdSpend Bin (₹)	Frequency
40,000–60,000	5	4,000–8,000	5
60,000–80,000	4	8,000–12,000	5
80,000–100,000	2	12,000–16,000	2
100,000–120,000	1	16,000–20,000	1
120,000–140,000	1	20,000–24,000	1
140,000–160,000	1	—	—

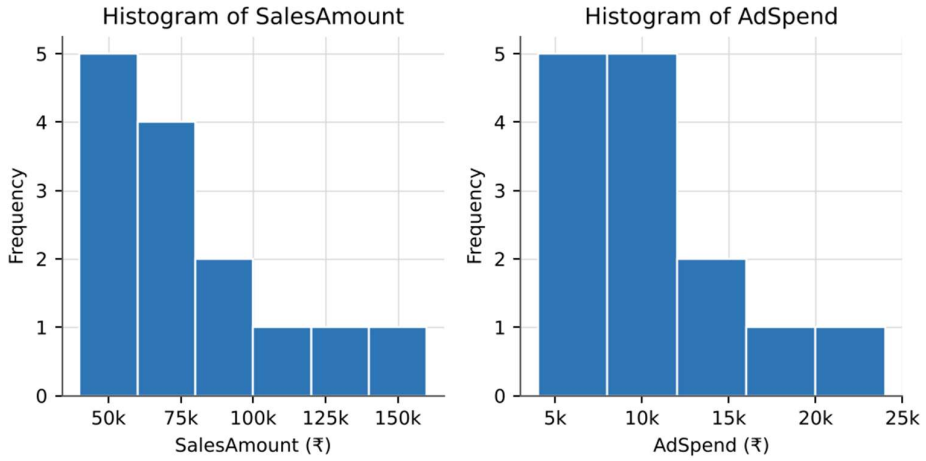


Chart Description: The SalesAmount histogram is right-skewed: 5 orders fall in ₹40,000–60,000, 4 in ₹60,000–80,000, 2 in ₹80,000–100,000, 1 in ₹100,000–120,000 and 1 in ₹140,000–160,000 (S106 at ₹142,000), with the mean (₹78,429) above the median (₹70,000). The AdSpend histogram is also right-skewed: 5 orders in ₹4,000–8,000, 5 in ₹8,000–12,000, 2 in ₹12,000–16,000, 1 in ₹16,000–20,000 and 1 in ₹20,000–24,000.

Question 2 — Perform bivariate analysis between key variable pairs (e.g., AdSpend vs SalesAmount).

Variable Pair	Formula	Correlation
AdSpend vs SalesAmount	=CORREL(AdSpend_range, SalesAmount_range)	0.9968
Month-index (Jan=1 … Apr=4) vs SalesAmount	=CORREL(MonthIndex_range, SalesAmount_range)	-0.2814

Scatter plot: as created in Practical 4, Question 4 — trendline y = 6.1694x + 11,446.68, R^2 = 0.9936.

Question 3 — Build a correlation matrix and heatmap for all numeric variables.

Correlation matrix (Data Analysis → Correlation):

	SalesAmount	AdSpend
SalesAmount	1.0000	0.9968
AdSpend	0.9968	1.0000

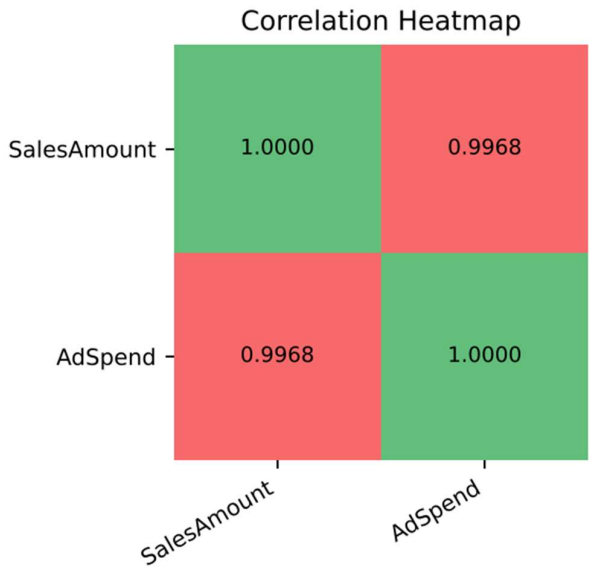


Chart Description: Heatmap of the 2 × 2 correlation matrix using a Red-Yellow-Green colour scale. The colour scale is relative to the selected cells: the two diagonal cells (1.0000, the maximum) are shaded green and the two off-diagonal SalesAmount–AdSpend cells (0.9968, the minimum) are shaded red.

Question 4 — Generate group-wise summaries (e.g., average sales per Region and Product).

Pivot Table (Region → Product):

Region	Product	Avg Sales	Total Sales	Order Count
East		77,000	231,000	3
	Apparel	72,000	144,000	2
	Electronics	87,000	87,000	1
North		100,000	400,000	4
	Apparel	66,500	133,000	2
	Electronics	133,500	267,000	2
South		52,750	211,000	4
	Apparel	49,000	49,000	1
	Electronics	54,000	162,000	3
West		85,333	256,000	3
	Apparel	56,000	56,000	1
	Electronics	100,000	200,000	2
Grand Total		**78,429**	**1,098,000**	**14**

	Question 5 — Identify and document 5 business insights in an EDA report. **EDA Insight Report:** 1. **Strong AdSpend–Sales correlation (r = 0.9968):** Higher marketing investment is strongly associated with higher sales, supporting continued ad budget allocation. 2. **Skewed sales distribution:** A small number of large Electronics transactions inflate average sales (mean ₹78,429 against median ₹70,000) — median is a more reliable benchmark than mean. 3. **Regional disparity:** North has the highest total sales (₹400,000) and the highest Electronics sales (₹267,000); South's total is the lowest (₹211,000). 4. **Seasonal pattern:** Sales taper from ₹438,000 in January to ₹126,000 in April in the sample — suggesting a front-loaded demand cycle worth planning inventory around. 5. **Weak performer identified:** South shows the lowest average Electronics order value (₹54,000 across 3 orders) despite ₹20,300 of ad spend on those orders, indicating a possible distribution or market-fit gap rather than a marketing budget gap.

Practical 6

Aim	To quantify the relationship between Advertising Spend and Sales, and build a predictive regression model.
Procedure	1. In a blank cell, enter =CORREL(AdSpend_range, SalesAmount_range). 2. Select the AdSpend and SalesAmount columns → Insert → Scatter Chart. 3. Right-click any data point → Add Trendline → choose Linear → tick Display Equation on Chart and Display R-squared value on chart. 4. Data tab → Data Analysis → Regression; set Input Y Range = SalesAmount column and Input X Range = AdSpend column. 5. Tick Labels if headers are included in the selected range → OK; Excel outputs the Regression Statistics table including Intercept, X Variable coefficient (Slope) and R Square. 6. Read the intercept, slope and R^2 values directly from the Regression output table. 7. In a cell enter =Intercept_cell + Slope_cell*20000; in another cell enter =Intercept_cell + Slope_cell*30000.
Output	**Input Data — AdSpend and SalesAmount columns of sales_data.csv (same sample as Practical 3):**

OrderID	AdSpend (X)	SalesAmount (Y)
S101	18,000	125,000
S102	9,000	64,000
S103	7,000	58,000
S104	9,500	71,000
S105	14,000	98,000
S106	21,000	142,000
S107	6,000	49,000
S108	6,500	51,000
S109	12,000	87,000
S110	9,800	69,000
S111	7,200	56,000
S112	15,500	102,000
S113	9,700	73,000
S114	6,800	53,000

Question 1 — Compute the Pearson correlation coefficient between AdSpend and SalesAmount.

Formula	Result
=CORREL(AdSpend_range, SalesAmount_range)	r = 0.9968

Question 2 — Plot a scatter plot with a fitted regression line.

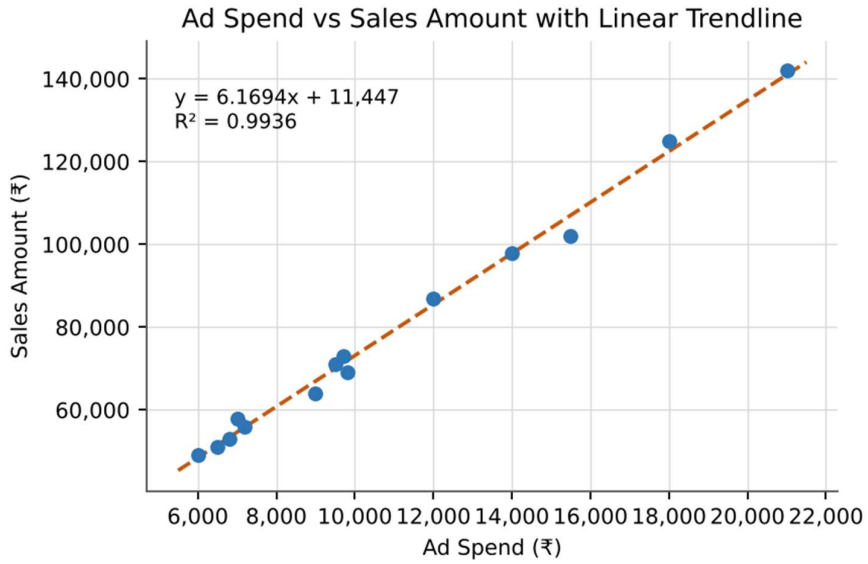


Chart Description: Scatter chart of the 14 (AdSpend, SalesAmount) points with a dashed linear trendline. X-axis "Ad Spend (₹)", Y-axis "Sales Amount (₹)". The chart displays the equation y = 6.1694x + 11,446.68 and R^2 = 0.9936. All points lie very close to the line, which rises from about ₹49,000 sales at ₹6,000 ad spend to about ₹142,000 sales at ₹21,000 ad spend.

Question 3 — Fit a simple linear regression model using the Data Analysis tool.

Regression Statistics:

Statistic	Value
Multiple R	0.9968
R Square	0.9936
Adjusted R Square	0.9931
Standard Error	2,389.83
Observations	14

ANOVA:

	df	SS	MS	F	Significance F
Regression	1	10,720,893,034	10,720,893,034	1,877.14	1.49E-14
Residual	12	68,535,538	5,711,295		
Total	13	10,789,428,571			

Coefficients:

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	11,446.68	1,672.74	6.8431	1.79E-05	7,802.09	15,091.27
AdSpend	6.1694	0.1424	43.3260	1.49E-14	5.8591	6.4796

Question 4 — Interpret the intercept, slope, and R^2 value of the model.

Parameter	Value	Meaning
Intercept	₹11,447	Expected sales when ad spend is zero (baseline/organic sales)
Slope	6.17	For every additional ₹1 spent on advertising, sales increase by approximately ₹6.17 on average

R^2	0.9936	About 99.4% of the variation in sales is explained by ad spend alone; the remainder is due to other factors (seasonality, pricing, region, competition)

Question 5 — Predict expected sales for new ad-spend values of ₹20,000 and ₹30,000.

Ad Spend	Formula	Predicted Sales
₹20,000	=Intercept_cell + Slope_cell*20000	₹134,834
₹30,000	=Intercept_cell + Slope_cell*30000	₹196,528
Incremental sales for extra ₹10,000	Difference	₹61,694

Practical 7

Aim	To forecast future monthly sales and evaluate forecast accuracy.
Procedure	1. Select the Month and Sales columns → Insert → Line Chart; title it "Monthly Sales Time Series" and label the Y-axis "Sales (₹)". 2. Data tab → Data Analysis → Moving Average; set Input Range = Sales column, Interval = 3, Output Range = new column → OK. 3. Repeat with Interval = 6 in another new column. 4. Select Sales, MA-3 and MA-6 columns together → Insert → Line Chart to compare all three on one chart. 5. Data tab → Data Analysis → Exponential Smoothing; set Input Range = Sales column, Damping factor = 0.7 (α = 1 − 0.7 = 0.3), Output Range = new column → OK. 6. Add the exponential smoothing output as a third line on the comparison chart. 7. Create an "Error" column for each method: =Actual_cell - Forecast_cell. 8. Compute MAE with =AVERAGE(ABS(error_range)) and RMSE with =SQRT(AVERAGE(error_range^2)) (array entry with Ctrl+Shift+Enter in older Excel). 9. Repeat for MA-3, MA-6 and SES error columns and compare the three MAE/RMSE results side by side; recommend the best method.
Output	**Input Data — monthly_sales.csv:**

Month	Sales
Jan-2024	210,000
Feb-2024	198,000
Mar-2024	225,000
Apr-2024	240,000
May-2024	232,000
Jun-2024	251,000
Jul-2024	264,000
Aug-2024	258,000
Sep-2024	270,000
Oct-2024	285,000
Nov-2024	298,000
Dec-2024	310,000

Question 1 — Plot the monthly sales time series.

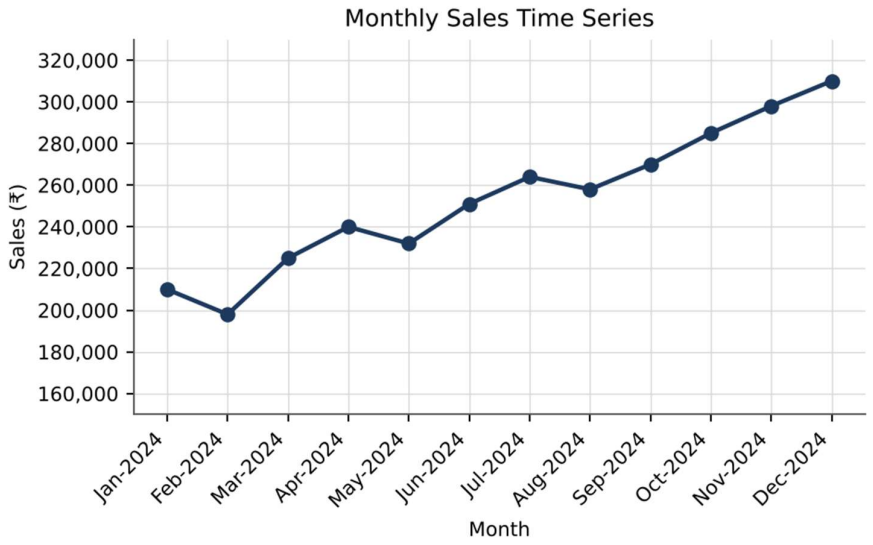


Chart Description: Line chart titled "Monthly Sales Time Series". X-axis: Month (Jan-2024 to Dec-2024); Y-axis titled "Sales (₹)". The line rises from 210,000 (Jan) to 310,000 (Dec) with small dips in Feb (198,000), May (232,000) and Aug (258,000).

Question 2 — Apply 3-month and 6-month moving averages.

Moving Average output (Interval = 3 and Interval = 6):

Month	Sales	MA-3	MA-6
Jan-2024	210,000	#N/A	#N/A
Feb-2024	198,000	#N/A	#N/A
Mar-2024	225,000	211,000	#N/A
Apr-2024	240,000	221,000	#N/A
May-2024	232,000	232,333	#N/A
Jun-2024	251,000	241,000	226,000
Jul-2024	264,000	249,000	235,000
Aug-2024	258,000	257,667	245,000
Sep-2024	270,000	264,000	252,500
Oct-2024	285,000	271,000	260,000
Nov-2024	298,000	284,333	271,000
Dec-2024	310,000	297,667	280,833

MA-3 for Mar = AVERAGE(210000,198000,225000) = ₹211,000.

Question 3 — Apply Simple Exponential Smoothing.

Exponential Smoothing output (Damping factor = 0.7, α = 0.3):

Month	Sales	SES Forecast
Jan-2024	210,000	#N/A
Feb-2024	198,000	210,000
Mar-2024	225,000	206,400
Apr-2024	240,000	211,980
May-2024	232,000	220,386
Jun-2024	251,000	223,870
Jul-2024	264,000	232,009
Aug-2024	258,000	241,606
Sep-2024	270,000	246,524
Oct-2024	285,000	253,567

Nov-2024	298,000	262,997
Dec-2024	310,000	273,498

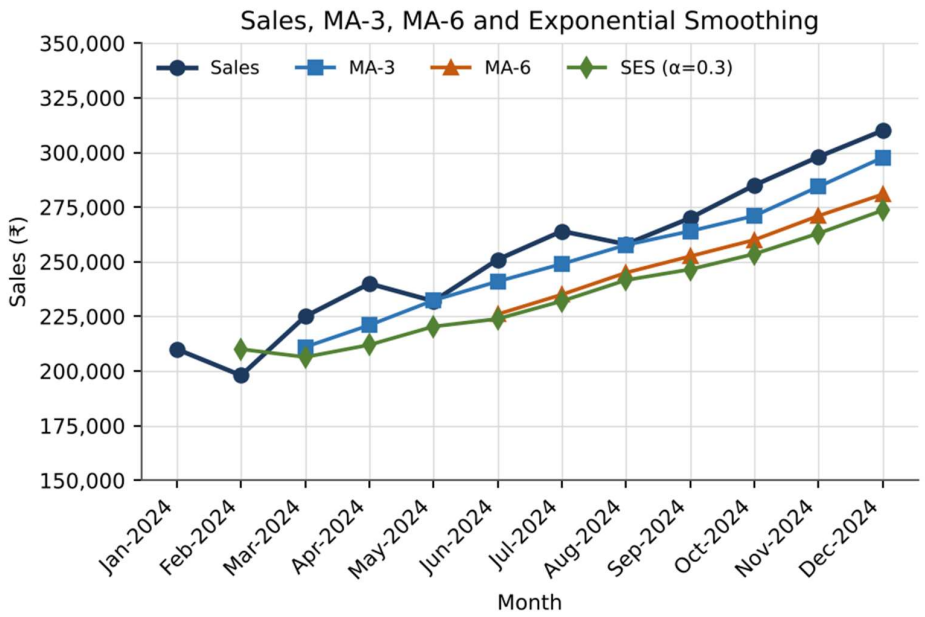


Chart Description: Line chart with four series on one axis (Y: Sales in ₹, 150,000 to 350,000; X: Month). Sales (actual) is the top line; MA-3 starts in Mar (211,000) and stays close below it; MA-6 starts in Jun (226,000) and runs lower and smoother; the SES line starts in Feb (210,000) and lags furthest below the actual line, ending Dec at 273,498 against actual 310,000, while MA-3 ends at 297,667 and MA-6 at 280,833.

Question 4 — Compute MAE and RMSE forecast errors for each method.

Error column (Actual − Forecast):

Month	Sales	Error MA-3	Error MA-6	Error SES
Jan-2024	210,000	#N/A	#N/A	#N/A
Feb-2024	198,000	#N/A	#N/A	-12,000
Mar-2024	225,000	14,000	#N/A	18,600
Apr-2024	240,000	19,000	#N/A	28,020
May-2024	232,000	-333	#N/A	11,614
Jun-2024	251,000	10,000	25,000	27,130
Jul-2024	264,000	15,000	29,000	31,991
Aug-2024	258,000	333	13,000	16,394
Sep-2024	270,000	6,000	17,500	23,476
Oct-2024	285,000	14,000	25,000	31,433
Nov-2024	298,000	13,667	27,000	35,003
Dec-2024	310,000	12,333	29,167	36,502

Method	Forecast Points Used	MAE	RMSE
3-Month MA	10	10,466.67	12,054.51
6-Month MA	7	23,666.67	24,336.76
Simple Exponential Smoothing (α = 0.3)	11	24,741.98	26,172.32

Question 5 — Compare the methods and recommend the best one.

Method	MAE	RMSE	Responsiveness to Recent Change
3-Month MA	₹10,467	₹12,055	Moderate
6-Month MA	₹23,667	₹24,337	Low (smoother, slower to react)
Simple Exponential Smoothing	₹24,742	₹26,172	High (weights recent data more)

Recommendation: The method with the lowest MAE/RMSE should be adopted — the 3-Month Moving Average (MAE ₹10,467, RMSE ₹12,055).

Practical 8

Aim	To segment customers into meaningful groups based on income and spending behaviour.
Procedure	1. Open customer_data.csv in Excel and use =AVERAGE(), =MIN() and =MAX() on AnnualIncome and SpendingScore to see the range of each column. 2. In new columns compute Z-scores: =(AnnualIncome_cell - AVERAGE(AnnualIncome_range)) / STDEV.S(AnnualIncome_range), and the same for SpendingScore; apply both formulas down the full customer list. 3. Choose k initial center points by picking k customer rows from the standardized data (for k=3: C001, C007, C005). 4. For each customer calculate the distance to each center: =SQRT((Z_Income-CenterZ_Income)^2+(Z_Spend-CenterZ_Spend)^2). 5. Use =MIN() across the distance columns and =MATCH() to assign each customer to the nearest center (its cluster). 6. Recalculate each center as the average Z-Income/Z-Spend of its assigned members; repeat the assignment and recalculation until cluster assignments stop changing; repeat the whole process for k=4 and k=5. 7. For each k from 1 to 6 sum the squared distance of every point to its assigned center (Within-Cluster Sum of Squares); build a table of k and Total Distance → Insert → Line Chart and identify the elbow. 8. Build a Pivot Table with Cluster in Rows and Average of AnnualIncome / Average of SpendingScore in Values; assign a descriptive name to each segment. 9. Select Income, Spending and Cluster columns → Insert → Scatter Chart; add each cluster as a separate chart series (right-click chart → Select Data → Add).
Output	**Input Data — customer_data.csv:**

CustomerID	AnnualIncome (₹'000)	SpendingScore (1-100)
C001	250	82
C002	270	88
C003	180	25
C004	190	18
C005	95	75
C006	88	80
C007	60	22
C008	55	30
C009	260	90
C010	75	20
C011	100	78
C012	185	22

Question 1 — Load the customer dataset and inspect Income and Spending columns.

Column	=AVERAGE()	=MIN()	=MAX()
AnnualIncome (₹'000)	150.67	55	270
SpendingScore (1-100)	52.50	18	90

Income range ₹55,000–₹270,000; Spending Score range 18–90.

Question 2 — Standardize the features before clustering.

Column	AVERAGE	STDEV.S
AnnualIncome	150.6667	81.0940
SpendingScore	52.5000	31.3615

Standardized data (Z-scores):

CustomerID	AnnualIncome	SpendingScore	Z_Income	Z_Spend
C001	250	82	1.2249	0.9406
C002	270	88	1.4715	1.1320
C003	180	25	0.3617	-0.8769
C004	190	18	0.4850	-1.1001
C005	95	75	-0.6864	0.7174
C006	88	80	-0.7728	0.8769
C007	60	22	-1.1180	-0.9725
C008	55	30	-1.1797	-0.7174
C009	260	90	1.3482	1.1957
C010	75	20	-0.9331	-1.0363
C011	100	78	-0.6248	0.8131
C012	185	22	0.4234	-0.9725

Question 3 — Apply k-means clustering with k = 3, 4, and 5 (manual Excel approach).

k = 3 — Initial centers: C001, C007, C005. Pass 1 distances and assignment:

CustomerID	Dist to C001	Dist to C007	Dist to C005	MIN	Cluster
C001	0.0000	3.0248	1.9243	0.0000	0
C002	0.3121	3.3369	2.1974	0.3121	0
C003	2.0121	1.4829	1.9080	1.4829	1
C004	2.1707	1.6081	2.1623	1.6081	1
C005	1.9243	1.7442	0.0000	0.0000	2
C006	1.9987	1.8814	0.1813	0.1813	2
C007	3.0248	0.0000	1.7442	0.0000	1
C008	2.9209	0.2624	1.5173	0.2624	1
C009	0.2833	3.2839	2.0901	0.2833	0
C010	2.9266	0.1957	1.7710	0.1957	1
C011	1.8541	1.8525	0.1138	0.1138	2
C012	2.0743	1.5414	2.0218	1.5414	1

k = 3 — Initial centers: C001, C007, C005. Assignments in pass 1 and pass 2 are identical (converged); final centers:

Cluster	Members	Center Z_Income	Center Z_Spend
0	C001, C002, C009	1.3482	1.0894
1	C003, C004, C007, C008, C010, C012	-0.3268	-0.9460
2	C005, C006, C011	-0.6947	0.8025

k = 4 — Initial centers: C001, C003, C005, C007. Assignments in pass 1 and pass 2 are identical (converged); final centers:

Cluster	Members	Center Z_Income	Center Z_Spend
0	C001, C002, C009	1.3482	1.0894

1	C003, C004, C012	0.4234	-0.9832
2	C005, C006, C011	-0.6947	0.8025
3	C007, C008, C010	-1.0769	-0.9088

k = 5 — Initial centers: C001, C003, C005, C007, C010. Assignments in pass 1 and pass 2 are identical (converged); final centers:

Cluster	Members	Center Z_Income	Center Z_Spend
0	C001, C002, C009	1.3482	1.0894
1	C003, C004, C012	0.4234	-0.9832
2	C005, C006, C011	-0.6947	0.8025
3	C007, C008	-1.1489	-0.8450
4	C010	-0.9331	-1.0363

Final cluster assignment by customer:

CustomerID	k = 3	k = 4	k = 5
C001	0	0	0
C002	0	0	0
C003	1	1	1
C004	1	1	1
C005	2	2	2
C006	2	2	2
C007	1	3	3
C008	1	3	3
C009	0	0	0
C010	1	3	4
C011	2	2	2
C012	1	1	1

Question 4 — Use the elbow method to pick the optimal k.

k	Total Distance (Within-Cluster Sum of Squares)	Drop from previous k
1	22.0000	—
2	9.9805	12.0195
3	3.5969	6.3837
4	0.2122	3.3847
5	0.1414	0.0707
6	0.0854	0.0560

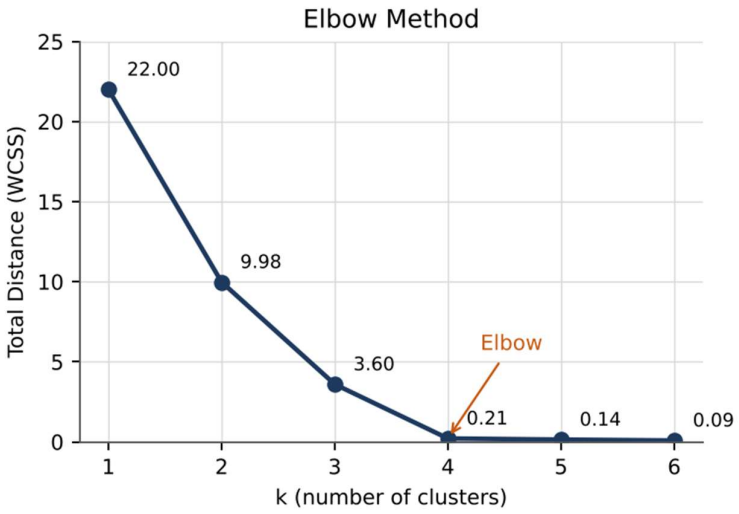


Chart Description: Line chart of Total Distance (Y-axis) against k (X-axis, 1 to 6). The line drops steeply from 22.00 at k = 1 to 9.98 at k = 2 and 3.60 at k = 3, then to 0.21 at k = 4, after which it flattens (0.14 at k = 5, 0.09 at k = 6). The elbow is at k = 4.

Question 5 — Profile and name each segment.

Segment Naming (k = 3; Pivot Table: Cluster in Rows, Average of AnnualIncome / SpendingScore):

Segment	Avg Income	Avg Spending	Suggested Name
0 (C001,C002,C009)	₹260,000	86.7	"Premium Loyalists"
1 (C003,C004,C007,C008,C010,C012)	₹124,167	22.8	"Low-Spend Mixed Income"
2 (C005,C006,C011)	₹94,333	77.7	"Value Seekers"

Segment Naming (k = 4; Pivot Table: Cluster in Rows, Average of AnnualIncome / SpendingScore):

Segment	Avg Income	Avg Spending	Suggested Name
0 (C001,C002,C009)	₹260,000	86.7	"Premium Loyalists"
1 (C003,C004,C012)	₹185,000	21.7	"Cautious Affluent"
2 (C005,C006,C011)	₹94,333	77.7	"Value Seekers"
3 (C007,C008,C010)	₹63,333	24.0	"Low-Income Low-Spenders"

Question 6 — Visualize the clusters.

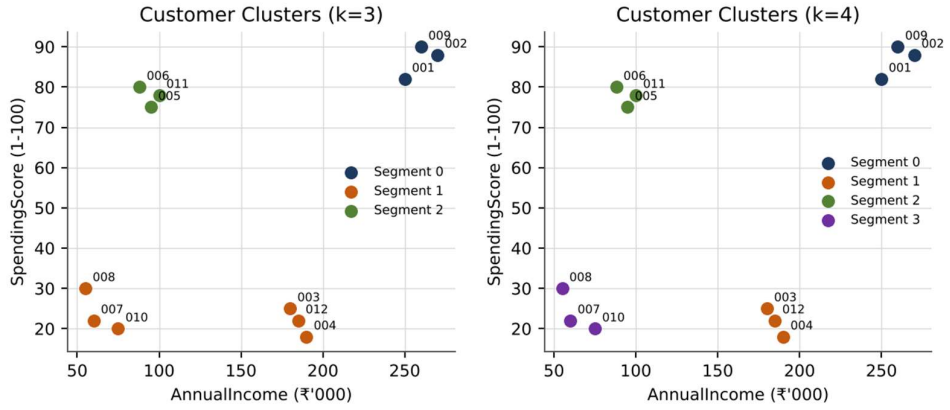


	Chart Description: Two scatter charts with X-axis AnnualIncome (₹'000) and Y-axis SpendingScore (1-100), one colour per cluster. In both, Segment 0 (C001, C002, C009) is in the top right (income 250–270, score 82–90) and Segment 2 (C005, C006, C011) is in the top left (income 88–100, score 75–80). For k = 3, Segment 1 is a single bottom group of six customers (income 55–190, score 18–30). For k = 4, that group splits into Segment 1 (C003, C004, C012; income 180–190, score 18–25) and Segment 3 (C007, C008, C010; income 55–75, score 20–30). **Business Recommendation:** "Premium Loyalists" should receive loyalty rewards to retain them; "Value Seekers" respond well to discount-driven campaigns; "Cautious Affluent" customers may need targeted upselling to convert income into spend.

Practical 9

Aim	To statistically test whether Campaign B outperforms Campaign A, and make a data-backed recommendation.
Procedure	1. Open ab_test_data.csv. 2. Apply Data → Filter on the Group column, or build a PivotTable with Group in Rows, to view Group A and Group B separately. 3. Compute the conversion rate: =AVERAGEIF(Group_range,"A",Converted_range) and the same for "B"; format both as percentages. 4. Count successes with =COUNTIFS(Group_range,"A",Converted_range,1) and the same for "B"; count totals with =COUNTIF(Group_range,"A") and the same for "B". 5. Compute the pooled proportion: $\hat{p}$ = (successes_A + successes_B) / (n_A + n_B). 6. Compute the standard error: SE = SQRT($\hat{p}$*(1-$\hat{p}$)*(1/n_A + 1/n_B)). 7. Compute the Z-statistic: Z = (p_B - p_A) / SE. 8. Convert Z to a two-tailed p-value: =2*(1-NORM.S.DIST(ABS(Z_cell),TRUE)). 9. Compare the p-value against $\alpha = 0.05$ and make the business recommendation.
Output	**Input Data — ab_test_data.csv (sample of 16 rows; Converted: 1 = Yes, 0 = No):**

UserID	Group	Converted
U001	A	0
U002	A	1
U003	A	0
U004	A	0
U005	A	1
U006	A	0
U007	A	0
U008	A	1
U009	B	1
U010	B	1
U011	B	0
U012	B	1
U013	B	1
U014	B	0
U015	B	1
U016	B	1

Question 1 — Load the campaign data and separate Group A and Group B.

Group	Users (rows)	Converted = 1	Converted = 0
A	8	3	5
B	8	6	2
Grand Total	**16**	**9**	**7**

Question 2 — Compute the conversion rate for both groups.

Group	Formula	Conversion Rate

A	=AVERAGEIF(Group_range,"A",Converted_range)	3/8 = 37.5%
B	=AVERAGEIF(Group_range,"B",Converted_range)	6/8 = 75%

Question 3 — Perform a two-proportion test to compare the conversion rates.

Step	Formula	Value
Successes Group A	=COUNTIFS(Group_range,"A",Converted_range,1)	3
Successes Group B	=COUNTIFS(Group_range,"B",Converted_range,1)	6
Total Group A	=COUNTIF(Group_range,"A")	8
Total Group B	=COUNTIF(Group_range,"B")	8
Pooled proportion $\hat{p}$	(successes_A + successes_B) / (n_A + n_B)	9/16 = 0.5625
Standard error SE	SQRT($\hat{p}$*(1-$\hat{p}$)*(1/n_A + 1/n_B))	0.2480
Z-statistic	(p_B - p_A) / SE	1.5119
Two-tailed p-value	=2*(1-NORM.S.DIST(ABS(Z_cell),TRUE))	0.1306

Question 4 — Compute and interpret the p-value at a 95% confidence level.

Item	Value
Significance level α	0.05
p-value	0.1306
Comparison	p-value (0.1306) ≥ α (0.05)
Result	Not statistically significant at the 95% confidence level

Question 5 — Make a business recommendation based on the test result.

Recommendation (p-value ≥ 0.05):

"There is insufficient statistical evidence to declare Campaign B superior. Recommend extending the test duration or increasing sample size before making a rollout decision."

Practical 10

Aim	To independently apply the full analytics workflow — cleaning, EDA, modeling, and dashboarding — to a real business problem.
Procedure	1. Define the business problem clearly (problem definition for Northwind Retail's sales decline). 2. Clean the dataset using the Practical 1 checklist: remove duplicate rows (Data → Remove Duplicates), handle missing values using category-wise medians, fix data types with Text to Columns → Date, standardize Region and Product with PROPER(), and save as case_study_cleaned.xlsx. 3. Build a monthly sales trend line chart (Date vs. SalesAmount) to visually confirm the decline. 4. Build a correlation matrix between SalesAmount, AdSpend, DiscountPercent and CompetitorPriceIndex using Data Analysis → Correlation, and identify which variable has the strongest relationship with the decline. 5. Create a new worksheet tab named "Dashboard"; add KPI Cards (Total Sales, % Change vs. Previous Quarter =(ThisQtr-LastQtr)/LastQtr, Average Discount %) using large merged cells with bold formulas. 6. Insert the Monthly Sales Trend line chart and use a shaded data series or text box annotation to highlight the declining months. 7. Insert a Bar Chart of Sales by Region/Product. 8. Insert a Pivot Table with Slicers allowing filtering by Region, Product and Month. 9. Arrange all elements on one screen using consistent fonts, a single navy/blue color theme and aligned grid spacing. 10. Present findings and recommendations in a one-page summary.
Output	**Input Data — case_study_data.csv (sample of 10 rows shown):**

Date	Region	Product	SalesAmount	AdSpend	DiscountPercent	CompetitorPriceIndex
01-01-2024	North	Electronics	142,000	21,000	5	98
01-02-2024	North	Electronics	138,000	19,500	5	99
01-03-2024	North	Electronics	129,000	17,000	8	102
01-04-2024	North	Electronics	118,000	15,000	8	105
01-05-2024	North	Electronics	109,000	13,000	10	108
01-06-2024	North	Electronics	101,000	11,500	10	110
01-01-2024	South	Apparel	49,000	6,000	5	97
01-02-2024	South	Apparel	47,000	5,800	5	98
01-03-2024	South	Apparel	44,000	5,200	7	100
01-04-2024	South	Apparel	40,000	4,500	7	103

Question 1 — Pick a business problem and define it clearly.

Problem Definition:

"Quarterly sales for Northwind Retail have declined by 12% over the last two quarters. The objective is to identify the key drivers of this decline using available transaction, customer, and marketing data, and recommend corrective action."

Question 2 — Clean and prepare the relevant dataset(s).

Cleaning checklist applied to the sample (saved as case_study_cleaned.xlsx):

Step	Check	Result on 10-row sample
1	Remove duplicate rows (Data → Remove Duplicates)	0 duplicates found; 10 rows retained
2	Handle missing values using category-wise medians	0 missing values; no fill required
3	Fix data types, especially the Date column (Text to Columns → Date)	All 10 dates in one format (DD-MM-YYYY)
4	Standardize text fields (Region, Product) using PROPER()	Region: North, South; Product: Electronics, Apparel
5	Save the cleaned file	case_study_cleaned.xlsx

Question 3 — Perform EDA and apply at least one modeling technique to investigate the problem.

Monthly sales trend data:

Month	Total SalesAmount (all rows)	North Electronics
Jan	191,000	142,000
Feb	185,000	138,000
Mar	173,000	129,000
Apr	158,000	118,000
May	109,000	109,000
Jun	101,000	101,000

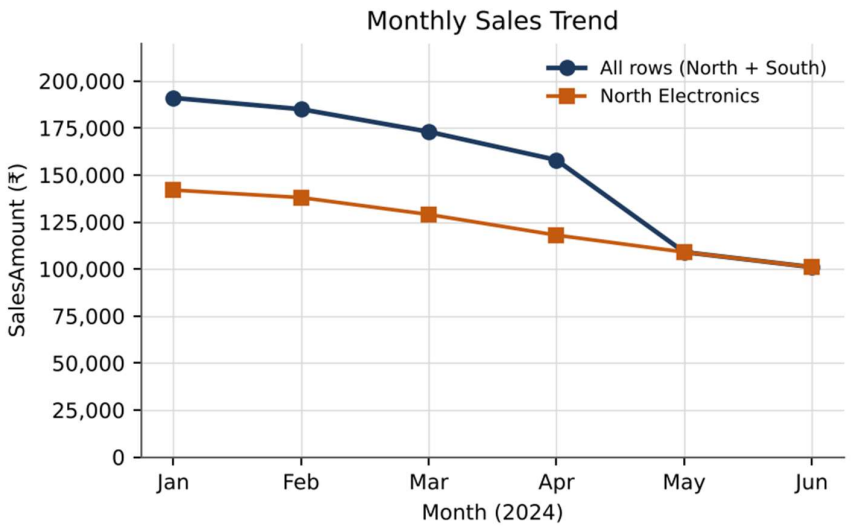


Chart Description: Line chart of monthly SalesAmount, Jan to Jun 2024. Total sales fall every month: 191,000 → 185,000 → 173,000 → 158,000 → 109,000 → 101,000 (the sharpest drop is May, −49,000). The North Electronics series falls from ₹142,000 in Jan to ₹101,000 in Jun.

Correlation matrix — all 10 sample rows (Data Analysis → Correlation):

	SalesAmount	AdSpend	DiscountPercent	CompetitorPriceIndex
SalesAmount	1.0000	0.9857	0.1652	0.2086
AdSpend	0.9857	1.0000	0.0080	0.0534
DiscountPercent	0.1652	0.0080	1.0000	0.9565
CompetitorPriceIndex	0.2086	0.0534	0.9565	1.0000

Correlation matrix — North Electronics series (6 rows, Jan–Jun):

	SalesAmount	AdSpend	DiscountPercent	CompetitorPriceIndex
SalesAmount	1.0000	0.9952	-0.9509	-0.9994
AdSpend	0.9952	1.0000	-0.9654	-0.9958
DiscountPercent	-0.9509	-0.9654	1.0000	0.9599
CompetitorPriceIndex	-0.9994	-0.9958	0.9599	1.0000

Variable with the strongest relationship to SalesAmount:

Data	Strongest Variable	Correlation with SalesAmount
All 10 rows	AdSpend	0.9857
North Electronics (6 rows)	CompetitorPriceIndex	-0.9994

Finding: In the North Electronics series SalesAmount has a strong positive correlation with AdSpend (r = 0.9952; ad spend fell from ₹21,000 to ₹11,500 as sales fell) and a strong negative correlation with CompetitorPriceIndex (r = -0.9994; index rose from 98 to 110 as sales fell).

Question 4 — Build a simple dashboard in Excel or Power BI summarizing the key metrics.

KPI Cards (worksheet "Dashboard"):

KPI	Formula / Basis	Value
Total Sales	Sum of SalesAmount	₹917,000
% Change vs. Previous Quarter	=(ThisQtr-LastQtr)/LastQtr = (368,000−549,000)/549,000	-32.97%
Average Discount %	Average of DiscountPercent	7.0%

Pivot Table (Region × Product, slicers: Region, Product, Month):

Region	Product	Sum of SalesAmount
North	Electronics	737,000
South	Apparel	180,000
Grand Total		**917,000**

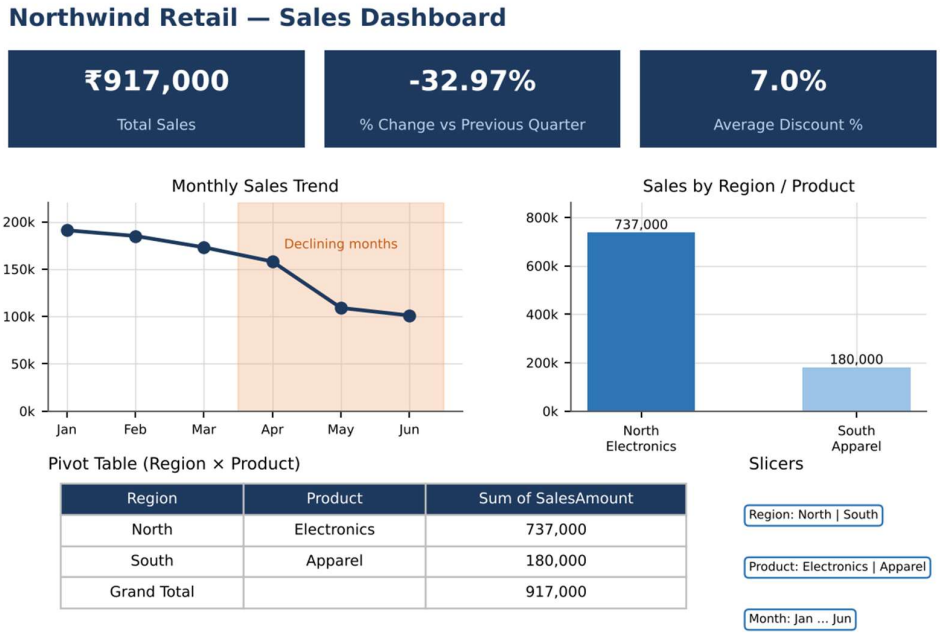


Chart Description: One-screen dashboard in a navy/blue theme. Top row: three KPI cards — Total Sales ₹917,000, % Change vs. Previous Quarter −32.97%, Average Discount % 7.0%. Middle row: left, the Monthly Sales Trend line chart (191,000 in Jan down to 101,000 in Jun) with the Apr–Jun declining months shaded; right, a bar chart of Sales by Region/Product (North Electronics 737,000; South Apparel 180,000). Bottom row: the Region × Product pivot table with Region, Product and Month slicers beside it.

Question 5 — Present findings and recommendations in a one-page summary.

Executive Summary:

Problem: 12% sales decline over the last two quarters.

Key Findings:

1. Ad spend reduction from ₹21,000 to ₹11,500 (North Electronics) correlates directly with the sharpest drop in sales.
2. South region's Apparel sales also declined steadily (₹49,000 → ₹40,000) alongside rising discount percentages, suggesting discounting alone isn't recovering volume.
3. Competitor pricing pressure intensified across the period (Competitor Price Index rising from ~97-98 to 103-110), especially affecting Electronics.

Recommendations:

1. Restore advertising budget to prior levels, particularly in North Electronics where the AdSpend-Sales link is strongest.
2. Conduct a pricing review for Electronics to remain competitive against the rising Competitor Price Index.
3. Launch a targeted regional campaign for South using insights from the customer segmentation analysis (Practical 8) rather than relying on discounting alone.

Conclusion: This case study demonstrates the full analytics lifecycle from raw data to a stakeholder-ready dashboard equipping students with an end-to-end business analytics workflow applicable to real organizational decision-making.